

# Database Design Example

COM 736 Database Systems and Data Analytics



# Scenario

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A University maintains information about the student registered for courses. Each student is registered for one course at a time. A course consists of a number of modules. A module is associated with one course only. For each student on a module a grade is awarded. Each module is led by one member of staff who belongs to a department. A department employs a number of staff. A department runs a number of courses.

# Data Requirements

- University may have different **course**, course details must be recorded
- Course have many **modules**, modules details must be stored
- **Staff** details and **Department** details must be recorded
- When a **student** register for a course, the details about the students and the course they are enrolled for must be recorded
- Students will be completing many modules as they take the course
  - For each module they will be awarded a grade

# Entities & Relationships

## Possible Entities

Student

Course

Module

Staff

Department

## Possible relationships

Student is registered for one course

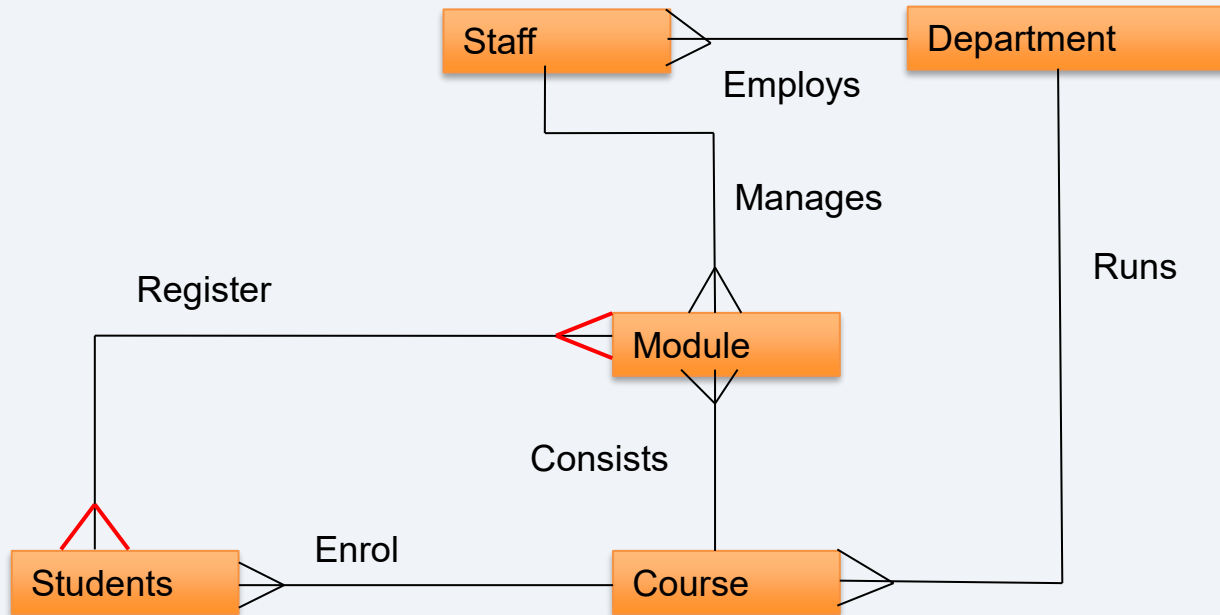
Course consists of a number of  
modules

Module is led by one member of staff

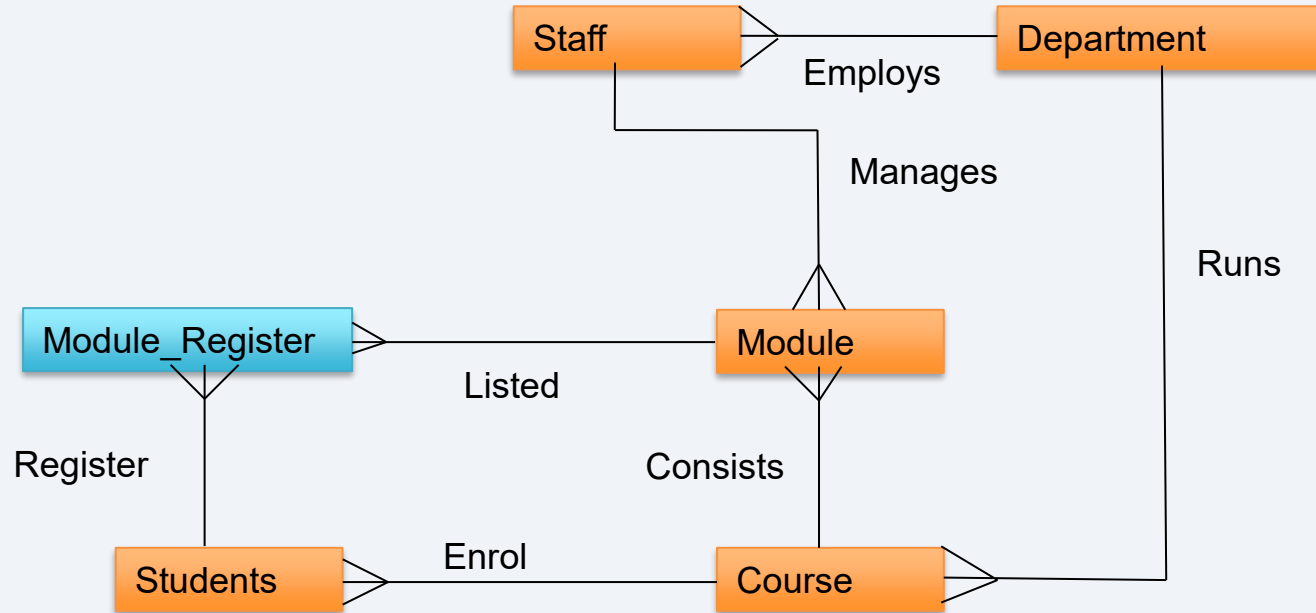
Department employs many staff

Department runs many courses

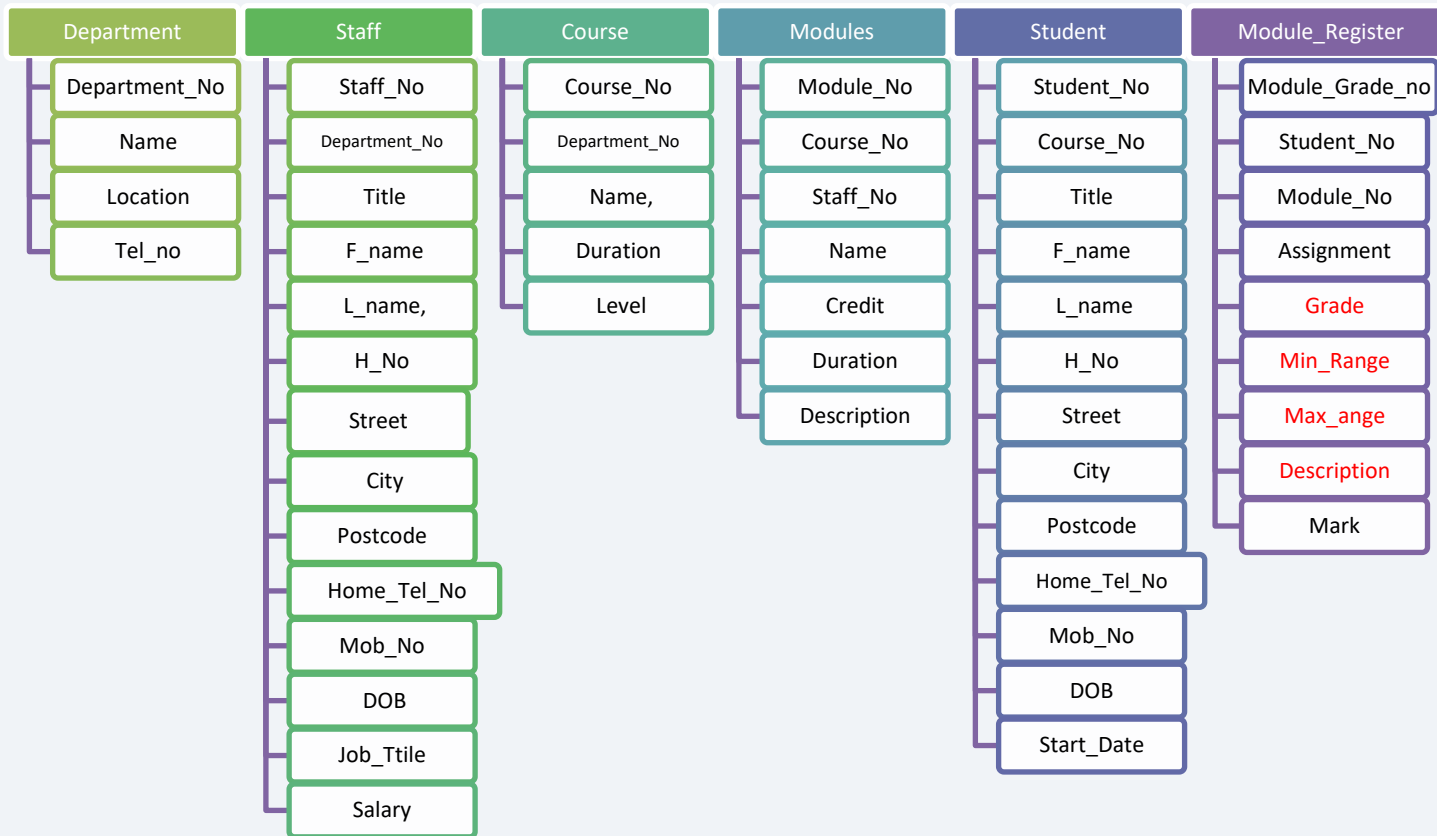
# Entities Relationship Diagram



# Entities Relationship Diagram - Solution



# Sample Attributes



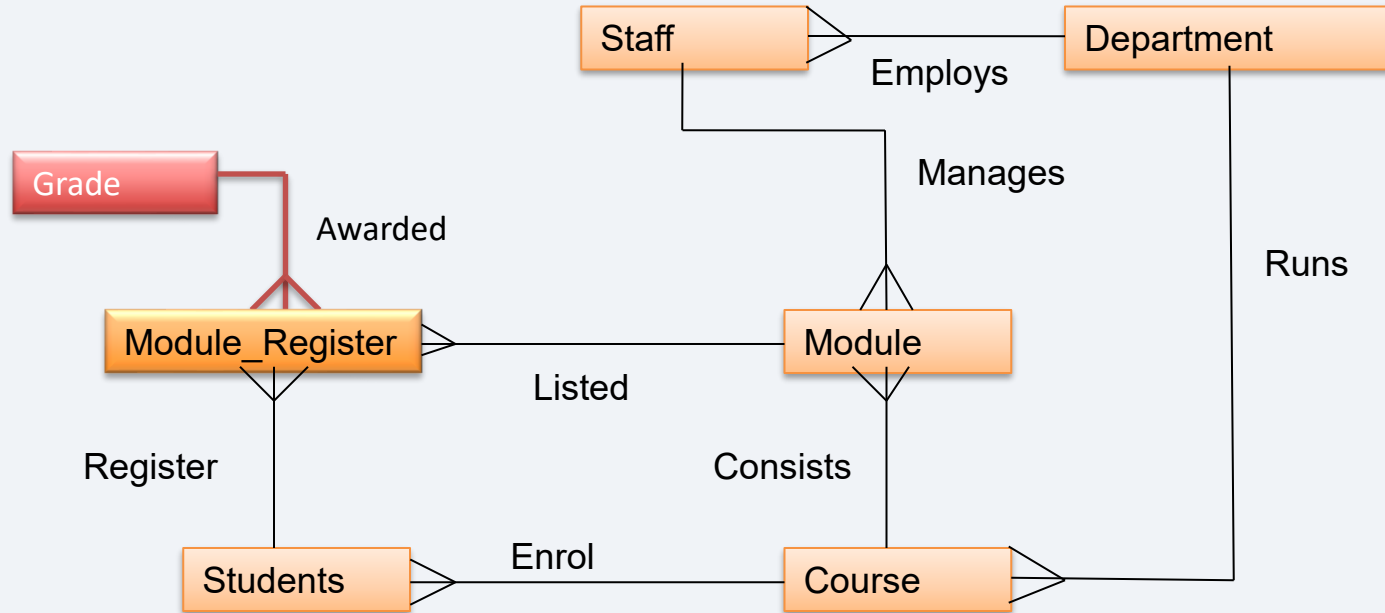
# Avoiding Data Duplication

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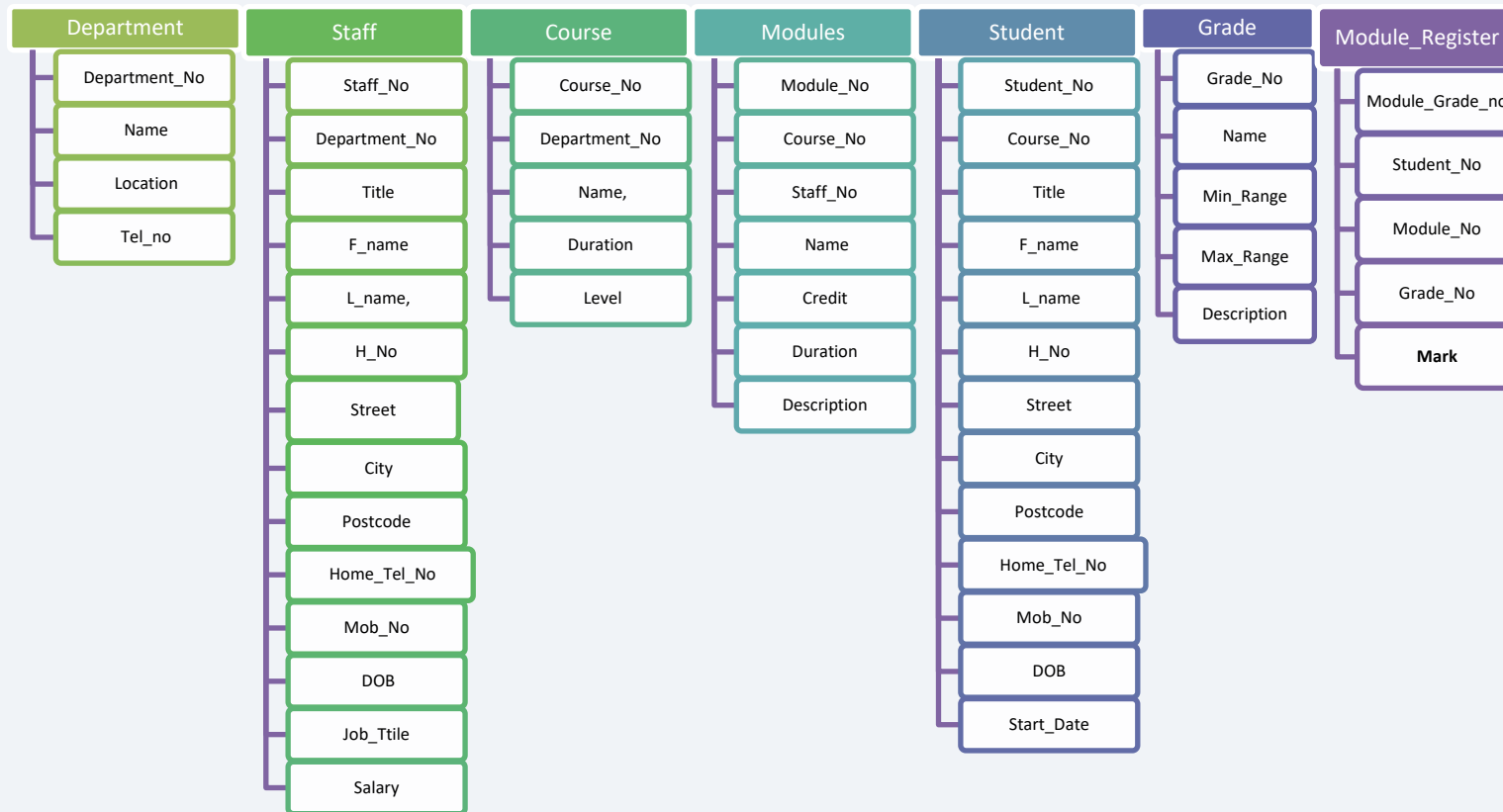
- Look closely on the Module Registration
  - Module\_Register
    - Module\_Grade\_no, Student\_No, Module\_No, Assignment, Grade, Min\_Range, Max\_Range, Description, Mark
  - Think about the Results/Grade
  - Grade Range and Description is repeated with every module registration.
  - Are there any better way of recording the data? Or avoiding the data duplication?



# Entities Relationship Diagram - Modified



# Sample Attributes



# Summary

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- Database design principles and stages
- Entity Relationship Model
  - Entities
  - Attributes
  - Relationship and cardinalities
- Entity Relationship Diagram
- Standards and notations